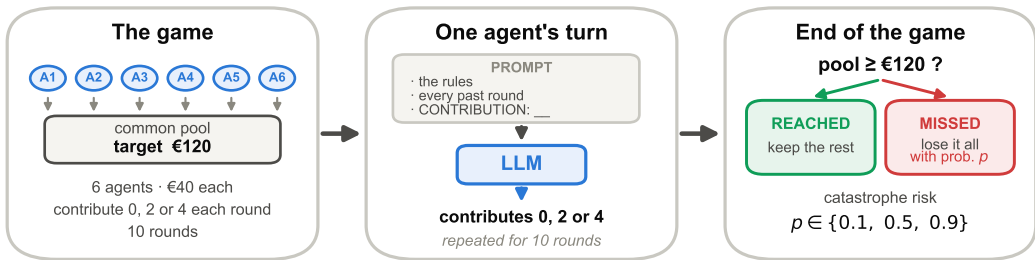


## (a) A collective-risk dilemma, played by six LLM agents



## (b) Two studies on the same games

### Study 1 · Behaviour

Do the agents cooperate more when the risk is higher, as humans do (Milinski et al. 2008)?

full factorial · 7B to frontier

13 models ×  $p = .1 / .5 / .9$  × EN · VN

= 60 games per configuration

human benchmark: 0% · 10% · 50% reach

measures: target-reach rate · how much they pool

### Study 2 · Comprehension

Is the deviation a real choice — or do they simply misunderstand the game?

the same prompt  
+ one probe question

graded against the  
engine's ground truth

Rules  
the fixed rules

Time  
read the history

State  
sum the pool

20 questions · 3 axes · ground truth from the engine

measure: answer accuracy (does it understand?)